

X200 Industrial Water Pump

Field Service & Troubleshooting Manual

Document ID: X200-FSM-001 **Revision:** 2.1 **Equipment:** X200 Pump

This sample manual is designed for a field-service troubleshooting system. It contains operating information, fault symptoms, likely causes, corrective actions, and service parts that can be used for retrieval and diagnosis.

1. Equipment Overview

The X200 is a centrifugal industrial water pump designed for continuous water circulation in treatment and process systems. The pump uses an electric motor, impeller, inlet filter, mechanical seal, discharge valve, and pressure sensor.

Normal operating conditions

Parameter	Normal Range
Discharge pressure	3.0–5.0 bar
Motor temperature	Below 80°C
Vibration	Below 4.5 mm/s RMS
Inlet flow	Minimum 40 L/min
Supply voltage	380–415 V AC

2. Common Faults

F-101 — Low discharge pressure

Symptoms: discharge pressure below 3.0 bar, reduced water flow, possible pump vibration.

Likely causes: clogged inlet filter, insufficient inlet water supply, damaged or worn impeller, partially closed discharge valve.

First action: inspect the inlet filter and confirm that the discharge valve is fully open.

Parts: IF-200 Inlet Filter; IMP-200 Impeller.

F-102 — Excessive vibration

Symptoms: vibration above 4.5 mm/s RMS, abnormal noise, unstable operation.

Likely causes: impeller imbalance, motor/pump misalignment, worn bearing, loose mounting bolts.

First action: stop the pump if vibration is severe; inspect mounting bolts and check alignment before restarting.

Parts: IMP-200 Impeller; BRG-200 Bearing Kit.

F-103 — Motor overheating

Symptoms: motor temperature above 80°C, thermal protection may activate.

Likely causes: blocked cooling path, excessive load, bearing friction, incorrect supply voltage.

First action: allow the motor to cool, inspect ventilation, and verify supply voltage.

Parts: BRG-200 Bearing Kit; FAN-200 Cooling Fan.

F-104 — Pump will not start

Symptoms: motor does not rotate when a start command is issued.

Likely causes: electrical supply interruption, motor protection trip, damaged contactor, control-system fault.

First action: verify electrical supply and check the motor protection status before attempting another start.

Parts: CNT-200 Contactor.

F-105 — Water leakage near shaft

Symptoms: visible water leakage around the pump shaft or seal housing.

Likely causes: worn mechanical seal, damaged seal surface, excessive shaft movement.

First action: shut down the pump and inspect the mechanical seal. Do not operate continuously with significant leakage.

Parts: SEAL-200 Mechanical Seal.

3. Detailed Troubleshooting Procedures

Low discharge pressure

1. Check inlet water supply is at least 40 L/min.
2. Inspect the inlet filter for blockage or debris.
3. Confirm the discharge valve is fully open.
4. If pressure remains below 3.0 bar, inspect the impeller for wear or damage.
5. Replace the inlet filter or impeller if inspection confirms damage.

Excessive vibration

1. Measure vibration at the motor and pump housing.
2. If vibration exceeds 4.5 mm/s RMS, inspect mounting bolts.
3. Check motor-to-pump alignment.
4. Inspect the impeller for damage or imbalance.
5. Inspect the bearing if vibration remains after alignment and impeller checks.

Motor overheating

1. Measure motor temperature.
2. Inspect cooling vents and fan for blockage.
3. Verify supply voltage is within 380–415 V AC.
4. Check for excessive mechanical load.
5. Inspect bearings if abnormal friction or noise is present.

Pump will not start

1. Verify incoming electrical supply.
2. Check whether motor protection has tripped.
3. Inspect the control-system start signal.
4. Inspect the contactor if supply is present but the motor does not energize.
5. Escalate to a qualified electrical technician if the control fault cannot be identified safely.

4. Service Parts Catalog

Part Number	Part	Used For
IF-200	Inlet Filter	Low pressure / restricted inlet flow
IMP-200	Impeller	Low pressure / vibration
BRG-200	Bearing Kit	Vibration / motor overheating
FAN-200	Cooling Fan	Motor overheating
CNT-200	Contactor	Pump will not start
SEAL-200	Mechanical Seal	Shaft-area water leakage

5. Safety and Escalation

Disconnect electrical power before opening electrical or rotating components. Do not operate the pump continuously when significant leakage, severe vibration, or abnormal temperature is observed. Electrical faults must be handled by qualified personnel.

If the available troubleshooting information does not identify a likely cause with sufficient evidence, escalate the case to a service engineer instead of replacing a component based only on a guess.

6. Diagnostic Reference Examples

Technician observation	Manual-supported interpretation
Pressure 2.2 bar + reduced flow	Check inlet filter, water supply, discharge valve, then impeller.
Vibration 6.1 mm/s + abnormal noise	Inspect mounting, alignment, impeller, and bearing.
Motor temperature 91°C	Inspect cooling, voltage, load, and bearing condition.
Water leakage near shaft	Inspect mechanical seal; shut down if leakage is significant.